

Turn this assignment in on gradescope. All work must be neat and legible. Turn in a "final draft". There should be nothing scratched out and your work should flow generally from left to right and top to bottom.

1. Compute 15 times 17 using the doubling algorithm.

'1 17

'2 34

'4 68

'8 136

Total: 15 255

2. Compute $180 \div 12$ using the doubling algorithm.

This is equivalent to "multiply 12 by what to get 180?"

	1	12'
	2	24'
	4	48'
	8	96'
Total:	15	180

Thus, $180 \div 12 = 15$.

3. Solve $\frac{7}{5}x + 3 = 24$ using false position.

$$\text{Guess } x = 5 : \frac{7}{5}(5) + 3 = 10 \quad (\text{Not high enough})$$

$$\text{Guess } x = 10 : \frac{7}{5}(10) + 3 = 17 \quad (\text{Still not high enough})$$

$$\text{Guess } x = 15 : \frac{7}{5}(15) + 3 = 24 \quad (\text{Correct})$$

4. Solve $\frac{6}{5}x + 3 = 20$ using false position.

$$\text{Guess } x = 5 : \frac{6}{5}(5) + 3 = 9 \quad (\text{Not high enough})$$

$$\text{Guess } x = 10 : \frac{6}{5}(10) + 3 = 15 \quad (\text{Still not high enough})$$

$$\text{Guess } x = 15 : \frac{6}{5}(15) + 3 = 21 \quad (\text{Too high})$$

$$\text{Guess } x = 14 : \frac{6}{5}(14) + 3 = 19 + \frac{4}{5} \quad \left(\text{Not high enough, need } \frac{1}{5} \right)$$

$$\text{Guess } x = 14 + \frac{1}{6} : \frac{6}{5} \left(14 + \frac{1}{6} \right) + 3 = 20 \quad (\text{Correct})$$